Retrospective validation of $P(v) = S_{\text{mutagen}}(v) \cdot \sigma(v)$ across three preclinical ALK TKI resistance screens

A Mutazyme II × alectinib (single-mutation anchor)

Variant	WT codon	Reachability	Class	Rate (%)	Chem-free rank	Simulated rank
V1180L	GTT	class-forced	C>G	5.0	8	15
L1196M	CTG	class-forced	C>A	18.0	1	1
I1171N	ATC	class-forced	T>A	10.5	61	118
I1171T	ATC	class-forced	T>C	18.0	22	10
L1198F	CTC	class-forced	C>T	47.0	18	3
G1202R	GGA	multi-class	—	—	—	—

B ePCR × neladalkib (compound-mutation, I1171N fixed-V1)

Variant	WT codon	Reachability	Class	Rate (%)	Chem-free rank	Simulated rank
L1256F	CTC	class-forced	C>T	47.0	124	44
E1129V	GAG	class-forced	T>A	10.5	642	726
E1129K	GAG	class-forced	C>T	47.0	854	266
C1156Y	TGC	class-forced	C>T	47.0	463	125
T1151R	ACG	class-forced	C>G	5.0	567	985

C ENU × neladalkib + alectinib + lorlatinib (industry screen, aggregate)

Variant	WT codon	Reachability	Class	Rate (%)	Chem-free rank	Simulated rank
I1171N	ATC	class-forced	T>A	31.4	61	40
I1171S	ATC	class-forced	T>G	5.0	26	34
I1171T	ATC	class-forced	T>C	36.6	22	6
F1174C	TTC	class-forced	T>G	5.0	5	10
F1174V	TTC	class-forced	T>G	5.0	16	20
V1180L	GTT	class-forced	C>G	**0.20**	8	229
T1151K	ACG	class-forced	C>A	9.0	72	165
T1151R	ACG	class-forced	C>G	**0.20**	104	1190
E1129K	GAG	class-forced	C>T	17.9	689	561
R1192P	?	unanchored	—	—	—	—
L1196M	CTG	class-forced	C>A	9.0	1	2
G1269A	GGA	class-forced	C>G	**0.20**	15	401
G1202R	GGA	multi-class	—	—	—	—